

Spectroscopy Electromagnet

Compact, Stable & Optical-Access Optimized
Dipole Electromagnet

**Precision Magnetic Fields.
Optical Measurement Access.
Built for Research Applications.**

The CRYONANO Spectroscopy Electromagnet is a compact, lightweight dipole electromagnet engineered for precision optical and spectroscopy-based research. Designed to deliver strong and stable magnetic fields within a minimal footprint, it integrates seamlessly with optical tables and laboratory setups.

Its low-profile coil architecture ensures unobstructed optical access, making it ideal for experiments requiring simultaneous magnetic field application and optical measurements.

KEY FEATURES

- **Compact & Lightweight Design**
Lightweight ~14 kg construction enables easy laboratory integration and handling.
- **High Magnetic Field Strength**
Provides excellent magnetic field stability for precision measurements.
- **Optical-Access Optimized Geometry**
Designed for enhanced optical accessibility in spectroscopy applications.
- **Adjustable Pole Gap**
Supports adjustable pole spacing from 0–40 mm for flexible setups.
- **Seamless Cryostat Integration**
Compatible with cryostat systems for low-temperature experiments.
- **Closed-Cycle Cooling System**
Enables continuous operation without liquid cryogen dependence.



Notes

- Designed for compact laboratory environments
- Optimized for simultaneous optical + magnetic measurements
- Ensures continuous operation with stable thermal performance
- Ideal where space, stability, and optical access are critical
- Supports integration with precision optical instrumentation

APPLICATIONS

- Optical spectroscopy experiments
- Laser-based measurements
- Magneto-optical studies like MOKE etc
- Semiconductor and photonics research
- Field-dependent material characterization
- Precision laboratory experiments with constraints

Spectroscopy Electromagnet

SPECIFICATIONS

MAGNET SPECIFICATION

Magnet Type	Compact Dipole Electromagnet
Pole Size	38 × 38 mm
Pole Gap	0 – 40 mm (adjustable)
Maximum Current	Up to 80 A
Magnetic Field	High intensity (application dependent)
Field Stability	High
Electrical Noise	Low ripple

MECHANICAL SPECIFICATION

Weight	~14 kg
Design	Low-profile, compact
Integration	Optical table compatible
Optical Access	Optimized, minimal obstruction

POWER SUPPLY SPECIFICATION

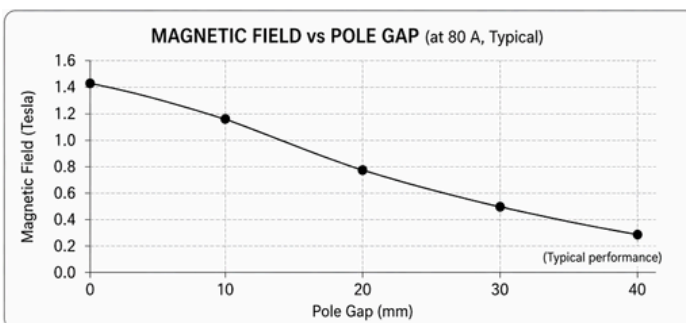
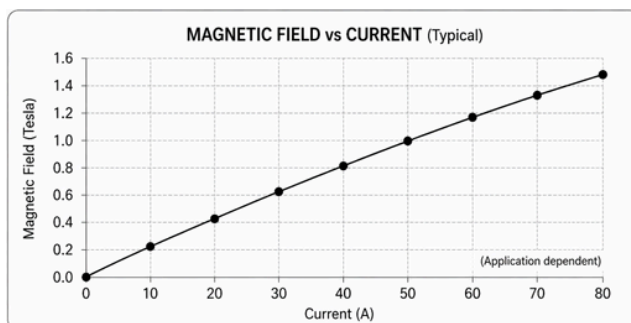
Type	SiC-based power supply
Power Rating	~1.5 kW
Output	Regulated DC
Operation	Stable, low-noise, unipolar

COOLING SPECIFICATION

Cooling Type	Closed-cycle cooling
Cooling Design	Copper–aluminium jacket
Operation	Continuous duty capable

COIL ARCHITECTURE

Design	Patent-pending low-height coil
Advantage	Enables optical access without compromising field uniformity
Thermal Performance	Optimized for efficient heat dissipation



Cryonano Labs | Head Office: Sector V, WTL Building, BP-5, BP Block, Sector V, Bidhannagar, Kolkata- 700091
Bangalore Office: #999/12, 2nd Floor, GMN Complex, 5th 'A' Cross, 1st Block, HRBR Layout, Babusapalya, Banaswadi,
Outer Ring Road, Bangalore – 560043.
www.cryonano.com | Phone: 097481 81485

Cryonano is not responsible for omissions or errors. Specifications subject to change without notice.
© 2026 CryoNano Instruments Pvt. Ltd. All rights reserved.